

READING PASSAGE 1

Quantum Entanglement and Information Theory

A The study of quantum entanglement and information theory has undergone a remarkable transformation over the past several decades. What was once a niche area of academic inquiry has developed into a rich and diverse field of research, drawing on insights from a wide range of disciplines and theoretical traditions.

B Historically, scholarly understanding of this topic was limited by both theoretical frameworks and available methodologies. Early researchers operated within relatively narrow conceptual boundaries, and their findings, while groundbreaking for their time, often failed to capture the full complexity of the phenomena they sought to explain. The development of new research tools and analytical techniques in the latter half of the twentieth century fundamentally transformed the landscape, enabling scholars to investigate questions that had previously been considered beyond the reach of empirical inquiry.

C At the core of current understanding lies a set of interrelated mechanisms that collectively explain the observed patterns and outcomes. First, structural factors create the underlying conditions within which individual and collective action takes place. These structural dimensions include economic incentives, institutional frameworks, and cultural norms that shape behaviour in ways that are often not immediately apparent to the actors themselves. Second, dynamic feedback processes amplify or dampen initial effects, generating patterns of change that can diverge substantially from what linear models would predict.

D The empirical record is further strengthened by a series of meta-analyses that have synthesised findings from hundreds of individual studies. These comprehensive reviews have consistently identified moderate to large effect sizes, with the most robust findings demonstrating remarkable consistency across different research designs, measurement approaches, and analytical strategies. This convergence of evidence from multiple independent lines of inquiry considerably strengthens confidence in the validity of the core theoretical propositions.

E Several unresolved debates continue to animate scholarly discussion in this field. Perhaps the most prominent concerns the relative importance of structural versus agentic factors in determining outcomes. While some researchers emphasise the constraining power of institutional and economic structures, others highlight the capacity of individuals and groups to exercise agency and effect change even within seemingly restrictive contexts. This debate has significant practical implications, as it shapes recommendations for policy interventions and institutional reforms.

F The future trajectory of research in this area will likely be shaped by several converging trends. Advances in computational methods and data science are opening up new analytical possibilities, enabling researchers to detect patterns and test hypotheses at scales that were previously unimaginable. Simultaneously, growing recognition of the importance of interdisciplinary collaboration is breaking down traditional barriers between fields and fostering innovative syntheses of diverse theoretical perspectives.

G In conclusion, the study of quantum entanglement and information theory represents a vital and dynamic area of contemporary research. The progress that has been made in understanding the key mechanisms and dynamics provides a solid foundation for both further scholarly inquiry and practical application. The challenges that remain should be seen not as reasons for discouragement but as opportunities for continued intellectual growth and social contribution.

Question 1

Choose the correct letter, A, B, C or D.

1 What finding emerged from the landmark study discussed in the passage?

- A. minimal impact
- B. creating underlying conditions for action
- C. determining individual choices directly
- D. irrelevant to outcomes

Question 2

Complete the sentences below. Choose NO MORE THAN THREE WORDS from the passage for each answer.

2 The author notes that researchers have called for more _____ approaches to investigation.

Question 3

Choose the correct letter, A, B, C or D.

3 Which of the following best describes the author's view on future developments?

- A. excessive research funding
- B. translating academic findings into practical policy
- C. lack of public interest
- D. too many international collaborations

Question 4

Do the following statements agree with the information given in the passage?

Write TRUE if the statement agrees with the information

Write FALSE if the statement contradicts the information

Write NOT GIVEN if there is no information on this

4 The landmark study discussed in the passage involved participants from multiple countries.

Question 5

Choose the correct letter, A, B, C or D.

5 What does the passage identify as a significant challenge to current understanding?

- A. excessive funding
- B. theoretical and methodological limitations
- C. public interest
- D. international collaboration

Question 6

Complete the sentences below. Choose NO MORE THAN THREE WORDS from the passage for each answer.

6 Researchers used data from more than _____ countries in the landmark study.

Question 7

Do the following statements agree with the information given in the passage?

Write TRUE if the statement agrees with the information

Write FALSE if the statement contradicts the information

Write NOT GIVEN if there is no information on this

- 7 Policy-makers have fully implemented all research recommendations.

Question 8

Choose the correct letter, A, B, C or D.

- 8 According to the text, what role do structural factors play?
- A. pessimism about the future
 - B. cautious optimism with acknowledgment of challenges
 - C. complete certainty about outcomes
 - D. indifference to practical applications

Question 9

Complete the sentences below. Choose NO MORE THAN THREE WORDS from the passage for each answer.

- 9 According to the passage, structural factors include economic incentives, institutional frameworks, and _____.

Question 10

Choose the correct letter, A, B, C or D.

- 10 What does the passage identify as a significant challenge to current understanding?
- A. weak effects across studies
 - B. moderate to large effect sizes with consistency
 - C. contradictory findings
 - D. results only in specific populations

Question 11

Complete the sentences below. Choose NO MORE THAN THREE WORDS from the passage for each answer.

- 11 Dynamic _____ processes can amplify or dampen initial effects in complex systems.

Questions 12–13

Do the following statements agree with the information given in the passage?

Write TRUE if the statement agrees with the information

Write FALSE if the statement contradicts the information

Write NOT GIVEN if there is no information on this

- 12 The landmark study discussed in the passage involved participants from multiple countries.

13 All critics agree that the research findings are universally applicable.